Slope, Motion & Comparing Rates

WORKSHEET | PART 1 OF 3 Questions 1–4

Show your reasoning and label any units.

Question 1 **R&P**

Position goes from 32 m at 0 s to 14 m at 6 s. Calculate average rate of change with units.

Question 2 **R&P**

Find the slope between (-2,7) and (4,-5). Show Δy and Δx.

Question 3 **R&P**

Two cyclists travel 30 m in 5 s and 42 m in 7 s. Compare their average rates.

Question 4 **R&P**

A tank starts at 70 gallons and loses 4 gallons per minute. Interpret the starting value and rate.

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WORKSHEET | PART 2 OF 3 Questions 5–8

Show your reasoning and label any units.

Question 5 **R&P**

Spiral: For {(0,5),(2,5),(2,7)}, is it a function? List domain/range.

Question 6 **R&P**

Preview: A scooter is 12 meters from a marker at time 0 and its position increases by 3 meters each second. Write an equation for position p after t seconds.

Question 7 **Matching practice**

An object is at 27 m after 1 s and 9 m after 7 s. Find the average rate of position change with units.

Question 8 **Matching practice**

Find the slope through (-1,-3) and (3,9), showing Δx and Δy.

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WORKSHEET | PART 3 OF 3 Questions 9–12

Show your reasoning and label any units.

Question 9 **Matching practice**

Cyclist A travels 45 m in 9 s. Cyclist B travels 40 m in 8 s. Compare their average speeds.

Question 10 **Matching practice**

A tank starts with 90 gallons and drains 6 gallons per minute. Interpret its starting value and rate of change.

Question 11 **Matching practice**

Is {(1,2), (1,4), (3,6)} a function? State the domain and range.

Question 12 **Matching practice**

An object starts 8 m from a marker and moves 4 m farther away each second. Write position p after t seconds.